

JAVA PROGRAMMING FOR BEGINNERS

25/03/25

Programming And Problem Solving

- I Love Programming

- You get to solve new problems every day
- Learn something new every day!

- Steps in problem Solving

- Step 1:- Understand the problem
- Step 2:- Design a Solution
 - Break The Problem into Small.
- Step 3:- Write your Program and Test
 - Express your Solution:- / language specifies Syntax

Let Solve multiple problem step by step

Learning the Program = learning to ride a bike

First Steps are the most difficult.
Pure fun afterwards

JShell

- * From Java 9:- Java is equally easy to learn - JShell.
- * Java REPL [Read Eval Print Loop]
- * JShell is a tool allows you to execute Java code getting immediate results.
- * Type in a one line of code and see the output.
- * If you type /exit $\rightarrow$ you will log out of JShell.

$\rightarrow$ $\times$ $\leftarrow$ $\times$ $\rightarrow$ $\times$ $\leftarrow$

The Challenge which we set ourselves is to Print the multiplication table.

Multiplication Table

Step by step

- How To break it down
- Where do we start
- Calculate 5×5
- Print $5 \times 5 = 25$
- Do This 10 Times

Pre Define Operator

Arithmetic Operators

* $\rightarrow$ Multiplication

/ $\rightarrow$ Division

% $\rightarrow$ Modulo

+ $\rightarrow$ Addition

- $\rightarrow$ Subtraction

* Expression

operator $\leftarrow$ $5 * 5$ $\rightarrow$ Operands or literals

An expression typically has operands and operator

Exercise

1. Write an expression to calculate number of Minutes in a day.
2. Write an expression to calculate number of seconds in a day

Solution

1. $60 * 24$ Ans:- 1440 min per day

2. $60 * 24 * 60$ Ans:- 86400 Sec per day

Puzzle with expressions

Example

1. $5 ** 5 \rightarrow$ illegal expressions

2. $5/2 \rightarrow 2$

- 5 is a integer number.
- 2 is also integer number. That y The result giving a 2.

• When you divide a integer by an integer, The result also integer. So 5 divided by 2 is 2

3. $5.0/2.0 \rightarrow 2.5$

- 5.0 is a decimal constant or floating constant
- 2.0 is a decimal constant or floating constant

4. $5.0/2 \rightarrow 2.5$

- 5.0 is a Floating point number.
- 2 is a integer number.
- When you divide a floating point by a integer, The result also get floating point.

$$5. \quad 5 + 5 * 6 \rightarrow 35$$

$$\begin{array}{r} 5 + 30 \\ \hline 35 \end{array}$$

In Programming Languages There is PRECEDENCE

PRECEDENCE

$$* / \% > + -$$

* / \% $\rightarrow$ These Three operators typically have greater precedence compared to plus and minus

Example:-

$1. \quad 5 - 2 * 2$ $\begin{array}{r} 5 - 4 \\ \hline 1 \end{array} \rightarrow 1$	$2. \quad 5 - 2 / 2$ $\begin{array}{r} 5 - 1 \\ \hline 4 \end{array} \rightarrow 4$
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If I want to do $5 - 2$ first and then multiply by 2. In programming languages we called as Brackets

Round Brackets [Parentheses]

We These () Brackets and put your expression between them

Example:-

$1. \quad (5 - 2) * 2$ $\begin{array}{r} 3 * 2 \\ \hline 6 \end{array} \rightarrow 6$	$2. \quad 5 - (2 * 2)$ $\begin{array}{r} 5 - 4 \\ \hline 1 \end{array}$
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Next Step - Printing output to console

Printing Text using `System.out.println`

`System.out.println(3*4);`

↓ output

12

→ integer value

Display a full equation

`System.out.println("5*2=10");`

↓ output

5*2=10

→ String value

Exercises

1. Print "Hello world"
2. Print $5*3$ as is
3. Print Calculated Values of $5*3$
4. How to Print Mathematical Table of 5 now?
5. Print number of Seconds in a day using `System.out.println`
6. Syntax Revision

Solution

1. `System.out.println("Hello world")` → Hello world
2. `System.out.println("5*3")` → 5*3
3. `System.out.println(5*3)` → 15
↳ parameter
4. `System.out.println("1*5=5")` → 1*5=5
↳ string parameter
5. `System.out.println(60*60*24)` → 86400
↳ parameter

Puzzle - Printing output to console

1. `System.out.println("Hello world")` → Hello world
2. * It ^{string} would print the Hello world with those many spaces between
2. `System.out.println(24*1*2)` → 48

* In expressions The spaces do not matter because it would print the value, The complete value of the expressions

3. `System.out.println("Helloworld")` → Helloworld

The next Thing case of the method

4. `System.out.println("Hello");` → error [Package System only ^{lower case} should capital other should be in lower case does not exist]
Syntax is a case sensitive

6. I need to print Double Quote in the output

1. `System.out.println("hello " world")`

↓ It says error

Solution.

• Escape character

* If I want to print Double Quote in the output
The way you can do that is by using something called
an escape character (\)

1. `System.out.println("hello \"world\")` → Hello "world"

2. `System.out.println("hello \"Varadth\"")` → hello "Varadth"

~~ES~~ \n
The next thing I would want to do split Hello and
World into two lines

* I can do that by using new line character \n

1. `System.out.println("Hello\nworld")` → Hello
world

~~IT~~
There is another escape character called \t
which is used to print a tab

1. `System.out.println("Hello\tworld")` → Hello world

* If I want to print \ itself then I would need to use \

1. `System.out.println("Hello \\ world")`
 → Hello \ world

* If I want to get two slashes → I ~~need~~^{should} do four slashes

1. `System.out.println("Hello \\\\ world")`
 → Hello \\ world.

Inbuilt Methods

• One of them used to do Mathematical Calculations

→ Syntax → Parameter.
 → `Math.random()` [For this we need to give parameter value]

* It generates a random value

* It does not expect any parameter.

* If you want to create a method or call a method, you need to use the brackets ()

1. `Math.random()` → 0.042407

This is a kind of random value. every time you get a different value.

→ `Math.min(5, 3)` → 3

• This syntax is used to find a minimum value of parameter.

• It says minimum of 5 and 3 is 3

→ Math. Max (23, 4) → 23

* This Syntax is used to find Maximum Values of Parameter.

* It Says Maximum of 23 and 4 is 23